

Que 1 →

- Create 2 Public Docker Hub registries named `cloudethix_master_nginx_yourname` & `cloudethix_release_nginx_yourname`.
- Clone below repository on your system.
`https://github.com/zembutsu/docker-sample-nginx.git`
- Initialize a local repository & copy the code from above repo to your local repository in master branch and then create below branches.
release
main
hotfix
- Once code is copied to local repository, from master branch update the `index.html` and add word "Cloudethix Master Branch Nginx" and build the docker image & add meaningful tags and push to Docker Hub registry `cloudethix_master_nginx_yourname`.
- Also from release branch update the `index.html` and add word "Cloudethix Release Branch Nginx" and build the docker image & add meaningful tags and push to Docker Hub registry `cloudethix_release_nginx_yourname`.
- Once Images are copied to Docker hub registries, switch to the main branch.
- In main branch create directory named `kube/clusterIP` & inside kube directory create file named `master_pod.yaml` with pod name `master_nginx` & with label `master_nginx` & add image that you have pushed in Docker Hub registry `cloudethix_master_nginx_yourname`.
- Also create a file `release_pod.yaml` with pod name `release_nginx` & with label `release_nginx` & add image that you have pushed in Docker Hub registry `cloudethix_release_nginx_yourname`.
- Create a file called `cluster_ip-service.yaml` with service name `cloudethix_clusterip` and with Type `clusterIP`.
- Then, select the pod with label `release_nginx` in service.
- Create all these three resources in your k8s cluster.
- Now, access `master_nginx` pod shell & curl the `master_nginx` pod & check the result.
- Also try to curl `release_nginx` pod with DNS name & check the result.
- Then curl the `clusterip` service with its name and check the result.
- Finally, create a GITHUB remote repository named `cloudethix-k8s-yourname` and push all the branches to the remote repository.
- Take all screenshots and create a well formatted document.

```
achal@LAPTOP-HPUOTJF1:Assignment-practice-2$ git clone
https://github.com/zembutsu/docker-sample-nginx.git
Cloning into 'docker-sample-nginx'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 22 (delta 7), reused 6 (delta 6), pack-reused 10 (from 1)
Receiving objects: 100% (22/22), done.
Resolving deltas: 100% (7/7), done.
```

```
achal@LAPTOP-HPUOTJF1:Assignment-practice-2$ cd Assignment-local-repo/
achal@LAPTOP-HPUOTJF1:Assignment-local-repo$ ls
docker-sample-nginx
achal@LAPTOP-HPUOTJF1:Assignment-local-repo$
```

```
achal@LAPTOP-HPUOTJF1:Assignment-local-repo$ git init
hint: Using 'master' as the name for the initial branch. This default branch
```

```
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in /home/achal/Assignment-practice-
2/Assignment-local-repo/.git/
achal@LAPTOP-HPU0TJF1:Assignment-local-repo$

-----

achal@LAPTOP-HPU0TJF1:Assignment-local-repo$ git branch release
achal@LAPTOP-HPU0TJF1:Assignment-local-repo$ git branch main
achal@LAPTOP-HPU0TJF1:Assignment-local-repo$ git branch hotfix
achal@LAPTOP-HPU0TJF1:Assignment-local-repo$ git branch
    hotfix
    main
* master
    release

-----

achal@LAPTOP-HPU0TJF1:clusterIp$ cat master_pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: master-nginx
  namespace: cloudethix-achal
  labels:
    app: master-nginx

spec:
  containers:
    - name: master-nginx
      image: docker.io/achalpadol/cloudethix_master_nginx_achal:master-v1
      ports:
        - containerPort: 80

-----

achal@LAPTOP-HPU0TJF1:clusterIp$ cat release_pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: release-nginx
  namespace: cloudethix-achal
  labels:
    app: release-nginx

spec:
  containers:
    - name: release-nginx
      image: docker.io/achalpadol/cloudethix_release_nginx_achal:release-v1
      ports:
        - containerPort: 80

-----

achal@LAPTOP-HPU0TJF1:clusterIp$ cat cluster_ip-service.yml
```

```
apiVersion: v1
kind: Service
metadata:
  name: cloudethix-clusterip
  namespace: cloudethix-achal
```

```
spec:
  type: clusterIp

  selector:
    app: release-nginx

  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
```

```
-----
achal@LAPTOP-HPUOTJF1:clusterIp$ kubectl get pod
NAME           READY   STATUS    RESTARTS   AGE
master-nginx   1/1     Running   0           3m37s
release-nginx  1/1     Running   0           2m26s
-----
```

```
achal@LAPTOP-HPUOTJF1:clusterIp$ kubectl apply -f cluster_ip-service.yml
service/cloudethix-clusterip created
```

```
achal@LAPTOP-HPUOTJF1:clusterIp$ kubectl get endpoints
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1
EndpointSlice
NAME                     ENDPOINTS              AGE
cloudethix-clusterip    10.244.2.107:80       11s
```

```
achal@LAPTOP-HPUOTJF1:clusterIp$ kubectl get svc
NAME                     TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
cloudethix-clusterip    ClusterIP   10.103.18.82  <none>       80/TCP     21s
-----
```

```
achal@LAPTOP-HPUOTJF1:clusterIp$ kubectl exec -it master-nginx -- /bin/sh
/ # curl http://10.244.1.21:80
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX Nginx</title>
</head>
<body>
  <h1>Cloudethix Master Branch Nginx</h1>
  <h2>Hostname: nginx</h2>
</body>
</html>
-----
```

```
/ # hostname
master-nginx
-----
```

```
achal@LAPTOP-HPUOTJF1:clusterIp$ kubectl exec -it release-nginx -- /bin/sh
/ # curl http://10.244.2.107:80
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX Nginx</title>
</head>
<body>
  <h1>Cloudethix Release Branch Nginx</h1>
```

```

    <h2>Hostname: nginx</h2>
</body>
</html>
-----

/ # hostname
release-nginx
-----

achal@LAPTOP-HPU0TJF1:clusterIp$ kubectl exec -it master-nginx -- /bin/sh
/ # curl http://cloudethix-clusterip
<!DOCTYPE html>
<html>
<head>
    <title>CloudEthiX Nginx</title>
</head>
<body>
    <h1>Cloudethix Release Branch Nginx</h1>
    <h2>Hostname: nginx</h2>
</body>
</html>
-----

achal@LAPTOP-HPU0TJF1:Assignment-local-repo$ git branch
    hotfix
* main
    master
    release
-----achal@LAPTOP-
HPU0TJF1:Assignment-local-repo$ git remote add origin
https://github.com/achalpadol/cloudethix-k8s-achalpadol.git
achal@LAPTOP-HPU0TJF1:Assignment-local-repo$ git remote -v
origin https://github.com/achalpadol/cloudethix-k8s-achalpadol.git (fetch)
origin https://github.com/achalpadol/cloudethix-k8s-achalpadol.git (push)
achal@LAPTOP-HPU0TJF1:Assignment-local-repo$ git push -u origin --all
Username for 'https://github.com': achalapdol
Password for 'https://achalapdol@github.com':
Enumerating objects: 2, done.
Counting objects: 100% (2/2), done.
Writing objects: 100% (2/2), 209 bytes | 209.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/achalpadol/cloudethix-k8s-achalpadol.git
 * [new branch]      hotfix -> hotfix
 * [new branch]      main -> main
 * [new branch]      master -> master
 * [new branch]      release -> release
branch 'hotfix' set up to track 'origin/hotfix'.
branch 'main' set up to track 'origin/main'.
branch 'master' set up to track 'origin/master'.
branch 'release' set up to track 'origin/release'.
-----

```

Que 2 →

- In the main branch of your local repository create a directory kube/NodePort.
- Create below files from below url. Please make sure you will create NodePort service with port 30008 instead of loadbalancer.
<https://kubernetes.io/docs/tasks/access-application-cluster/connecting-frontend-backend/>.
 - backend-deployment.yaml
 - backend-service.yaml
 - frontend-deployment.yaml
 - frontend-NodePort-service.yaml

- Once files are created , create all the resources in your k8s cluster.
 - Access all public ips with port 30008 in the browser and then check the result.
 - Finally, push all the latest code to the remote repository.
 - Take all screenshots and create a well formatted document
-

```
achal@LAPTOP-HPUOTJF1:NodePort$ cat backend-deployment.yaml backend-service.yaml frontend-NodePort-service.yaml frontend-deployment.yaml
```

```
apiVersion: apps/v1
```

```
kind: Deployment
```

```
metadata:
```

```
  name: backend
```

```
  namespace: cloudeithx-achal
```

```
spec:
```

```
  replicas: 3
```

```
  selector:
```

```
    matchLabels:
```

```
      app: hello
```

```
      tier: backend
```

```
      track: stable
```

```
  template:
```

```
    metadata:
```

```
      labels:
```

```
        app: hello
```

```
        tier: backend
```

```
        track: stable
```

```
    spec:
```

```
      containers:
```

```
        - name: hello
```

```
          image: gcr.io/google-samples/hello-go-gke:1.0
```

```
      ports:
```

```
        - name: http
```

```
          containerPort: 80
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: hello
```

```
  namespace: cloudeithx-achal
```

```
spec:
```

```
  selector:
```

```
    app: hello
```

```
    tier: backend
```

```
  ports:
```

```
    - protocol: TCP
```

```
      port: 80
```

```
      targetPort: http
```

```
apiVersion: v1
```

```
kind: Service
```

```
metadata:
```

```
  name: frontend
```

namespace: cloude**thix**-achal

spec:
type: NodePort

selector:
app: hello
tier: frontend

ports:
- protocol: TCP
port: 80
targetPort: 80
nodePort: 30008

apiVersion: apps/v1
kind: Deployment
metadata:
name: frontend
namespace: cloude**thix**-achal

spec:
replicas: 1

selector:
matchLabels:
app: hello
tier: frontend
track: stable

template:
metadata:
labels:
app: hello
tier: frontend
track: stable

spec:
containers:
- name: nginx
image: gcr.io/google-samples/hello-frontend:1.0

lifecycle:
preStop:
exec:
command:
- /usr/sbin/nginx
- -s
- quit

achal@LAPTOP-HPUOTJF1:NodePort\$ kubectl get pod

NAME	READY	STATUS	RESTARTS	AGE
backend-7cdc9fcc87-cbtrs	1/1	Running	0	18m
backend-7cdc9fcc87-dl977	1/1	Running	0	18m
backend-7cdc9fcc87-qtljg	1/1	Running	0	18m
frontend-7db75b888d-4ljtx	1/1	Running	0	6m15s

achal@LAPTOP-HPUOTJF1:NodePort\$ kubectl get svc

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
frontend	NodePort	10.103.159.53	<none>	80:30008/TCP	5m44s

hello ClusterIP 10.109.242.247 <none> 80/TCP 4m13s

achal@LAPTOP-HPUOTJF1:NodePort\$ minikube service frontend -n cloudehix-achal

NAMESPACE	NAME	TARGET PORT	URL
cloudehix-achal	frontend	80	http://192.168.49.2:30008

🔗 Starting tunnel for service frontend.

NAMESPACE	NAME	TARGET PORT	URL
cloudehix-achal	frontend		http://127.0.0.1:41693

🌐 Opening service cloudehix-achal/frontend in default browser...

👉 http://127.0.0.1:41693

! Because you are using a Docker driver on linux, the terminal needs to be open to run it.

achal@LAPTOP-HPUOTJF1:~\$ curl http://127.0.0.1:41693
{\"message\":\"Hello\"}
achal@LAPTOP-HPUOTJF1:~\$ curl http://192.168.49.2:30008
{\"message\":\"Hello\"}

achal@LAPTOP-HPUOTJF1:NodePort\$ git push
Username for 'https://github.com': achalpadol
Password for 'https://achalpadol@github.com':
Enumerating objects: 13, done.
Counting objects: 100% (13/13), done.
Delta compression using up to 12 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 1.62 KiB | 1.62 MiB/s, done.
Total 12 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/achalpadol/cloudehix-k8s-achalpadol.git
305bba0..05d67b5 main -> main

Que 3 →
• Create any 2 pods and assign them to different worker nodes with nodeName property.

achal@LAPTOP-HPUOTJF1:Node-selector-pod\$ cat pod-1.yml pod-2.yml
apiVersion: v1
kind: Pod
metadata:
 name: pod-1
 namespace: cloudehix-achal
spec:
 nodeName: minikube-m02
 containers:
 - name: my-containers
 image: nginx:latest
 ports:
 - containerPort: 80

apiVersion: v1
kind: Pod
metadata:

```
name: pod-2
namespace: cloudehix-achal
spec:
  nodeName: minikube-m03
  containers:
    - name: my-container-2
      image: nginx:latest
      ports:
        - containerPort: 80
```

```
achal@LAPTOP-HPUOTJF1:Node-selector-pod$ kubectl get pod -o wide
NAME      READY   STATUS    RESTARTS   AGE      IP             NODE
NOMINATED NODE   READINESS GATES
pod-1     1/1     Running   0           2m53s    10.244.1.24    minikube-m02
<none>    <none>
pod-2     1/1     Running   0           2m2s     10.244.2.109   minikube-m03
<none>    <none>
```

Que 4 →

- Label both worker nodes such as worker-0 node as cloudehix-k8s-00 & worker-1 node as cloudehix-k8s-01.
- Once nodes are labeled, create pod00.yaml file and schedule the pod on worker-0 node with nodeSelector property. Also create one more file named pod01.yaml & schedule the pod on worker-1 node.

```
achal@LAPTOP-HPUOTJF1:~$ kubectl label node minikube-m02 cloudehix-k8s-00=worker-0
node/minikube-m02 labeled
achal@LAPTOP-HPUOTJF1:~$ kubectl label node minikube-m03 cloudehix-k8s-00=worker-1
node/minikube-m03 labeled
achal@LAPTOP-HPUOTJF1:~$ kubectl get nodes --show-labels
NAME                STATUS    ROLES          AGE   VERSION   LABELS
minikube            Ready     control-plane   44h   v1.35.1   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube,kubernetes.io/os=linux,minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=true,minikube.k8s.io/updated_at=2026_09_03T12_36_51_0700,minikube.k8s.io/version=v1.38.1,node-role.kubernetes.io/control-plane=,node.kubernetes.io/exclude-from-external-load-balancers=
minikube-m02       Ready     <none>          44h   v1.35.1   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,cloudehix-k8s-00=worker-0,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube-m02,kubernetes.io/os=linux,minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=false,minikube.k8s.io/updated_at=2026_09_03T12_38_34_0700,minikube.k8s.io/version=v1.38.1
minikube-m03       Ready     <none>          44h   v1.35.1   beta.kubernetes.io/arch=amd64,beta.kubernetes.io/os=linux,cloudehix-k8s-00=worker-1,kubernetes.io/arch=amd64,kubernetes.io/hostname=minikube-m03,kubernetes.io/os=linux,minikube.k8s.io/commit=c93a4cb9311efc66b90d33ea03f75f2c4120e9b0,minikube.k8s.io/name=minikube,minikube.k8s.io/primary=false,minikube.k8s.io/updated_at=2026_09_03T12_44_45_0700,minikube.k8s.io/version=v1.38.1
```

Que 5 →

- Clone the below repo locally & create DaemonSet from directory DaemonSet101.

<https://github.com/collabnix/kubelab>

```
achal@LAPTOP-HPUOTJF1:daemonSet$ git clone
```



```
https://github.com/collabnix/kubelabs.git
Cloning into 'kubelabs'...
remote: Enumerating objects: 14031, done.
remote: Counting objects: 100% (707/707), done.
remote: Compressing objects: 100% (147/147), done.
remote: Total 14031 (delta 637), reused 560 (delta 560), pack-reused 13324
(from 5)
Receiving objects: 100% (14031/14031), 61.81 MiB | 1.04 MiB/s, done.
Resolving deltas: 100% (4181/4181), done.
```

```
-----
achal@LAPTOP-HPUOTJF1:DaemonSet101$ cat daemonset.yml
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: prometheus-daemonset
spec:
  selector:
    matchLabels:
      tier: monitoring
      name: prometheus-exporter
  template:
    metadata:
      labels:
        tier: monitoring
        name: prometheus-exporter
    spec:
      containers:
        - name: prometheus
          image: prom/node-exporter
          ports:
            - containerPort: 80
```

```
-----
achal@LAPTOP-HPUOTJF1:DaemonSet101$ kubectl apply -f daemonset.yml
daemonset.apps/prometheus-daemonset created
achal@LAPTOP-HPUOTJF1:DaemonSet101$ kubectl get daemonset
NAME                                DESIRED    CURRENT    READY    UP-TO-DATE    AVAILABLE
NODE SELECTOR  AGE
prometheus-daemonset  3          3          2        3             2
<none>           26s
```

```
-----
achal@LAPTOP-HPUOTJF1:static-pod$ kubectl get pod
NAME                                READY    STATUS    RESTARTS    AGE
prometheus-daemonset-gz2q8  1/1     Running   0            8m12s
prometheus-daemonset-mhn62  1/1     Running   0            8m12s
prometheus-daemonset-pbfm7  1/1     Running   0            8m12s
```

```
-----
Que 6 →
● Create a static pod with name cloudethix-static in your k8s cluster. Refer below link.
https://kubernetes.io/docs/tasks/configure-pod-container/static-pod
```

```
-----
achal@LAPTOP-HPUOTJF1:static-pod$ cat cloudethix-static-pod.yml
apiVersion: v1
kind: Pod
metadata:
  name: cloudethix-static
  namespace: cloudethix-achal
  labels:
    role: myrole
spec:
```

```
containers:
  - name: web
    image: nginx
    ports:
      - name: web
        containerPort: 80
        protocol: TCP
```

```
-----
achal@LAPTOP-HPUOTJF1:static-pod$ minikube cp cloudethix-static-pod.yml
/etc/kubernetes/manifests/cloudethix-static-pod.yml
achal@LAPTOP-HPUOTJF1:static-pod$ minikube ssh
Linux minikube 6.6.114.1-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Mon
Dec  1 20:46:23 UTC 2025 x86_64
```

The programs included with the Debian GNU/Linux system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright.

```
Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
docker@minikube:~$ ls /etc/kubernetes/manifests/
cloudethix-static-pod.yml  etcd.yaml  kube-apiserver.yaml  kube-controller-
manager.yaml  kube-scheduler.yaml
```

```
-----
achal@LAPTOP-HPUOTJF1:static-pod$ kubectl get pod -A
NAMESPACE          NAME                                READY   STATUS
RESTARTS    AGE
cloudethix-achal  cloudethix-static-minikube        1/1     Running   0
2m2s
```

```
-----
Que 7 →
● Install Kubectl & kubens in your k8s cluster
```

```
-----
sudo apt update
sudo apt install kubectl -y
```

```
achal@LAPTOP-HPUOTJF1:~$ sudo apt install kubectl -y
kubectl is already the newest version (0.9.5-2build1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 40
```

```
kubectl
kubens

kubectl minikube
kubens cloudethix-achal
-----
```